

17 Binomial Distributions

Example 17.1. Capture-recapture sampling is a technique often used to estimate the size of a population. Suppose you want to estimate N , the number of [monarch butterflies in Pismo Beach](#). (Assume that N is a fixed but unknown number; the population size doesn't change over time.) You first capture a sample of N_1 butterflies, selected randomly, and tag them and release them. At a later date, you then capture a second sample of n butterflies, selected randomly *with replacement*. Let X be the number of butterflies in the second sample that have tags (because they were also caught in the first sample). (Assume that the tagging has no effect on behavior, so that selection in the first sample is independent of selection in the second sample.)

In practice, N is unknown and the point of capture-recapture sampling is to estimate N . But let's start with a simpler, but unrealistic, example where there are $N = 52$ butterflies, $N_1 = 13$ are tagged and $N_0 = 52 - 13 = 39$ are not, and $n = 5$ is the size of the second sample.

1. Can the selections (for the second sample) be considered Bernoulli trials?
2. What are the possible values of X ?
3. Describe in detail how you could use simulation to approximate the distribution of X .
4. Compute and interpret $P(X = 0)$.

5. Compute the probability that in the second sample the first butterfly selected is tagged but the rest are not.
6. Explain why the previous part is not $P(X = 1)$.
7. How many outcomes (S/F sequences of length 5) satisfy the event $\{X = 1\}$? Answer without listing all the possibilities.
8. Compute and interpret $P(X = 1)$.
9. Compute and interpret $P(X = 2)$.
10. Suggest a general formula for the probability mass function of X . Then use the pmf to create a table and spinner representing the distribution of X .
11. Suggest a simple shortcut formula for $E(X)$.

12. Compute $E(X)$. Did the shortcut formula work?

13. Compute $\text{Var}(X)$.

- Consider a *fixed number*, n , of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
 - n (an integer): the fixed number of Bernoulli trials, or the sample size
 - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n , a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

- If X has a Binomial(n, p) distribution then

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1-p) \end{aligned}$$

Example 17.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p . If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.
3. Roll a die until it lands on 6; X is the total number of rolls.
4. Roll a die until it lands on 6 three times; X is the total number of rolls.
5. Roll a die 20 times; X is the sum of the numbers rolled.
6. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)
7. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

Example 17.3. Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z .
2. Are X and Z independent? Explain.

3. Compute $P(X = 2, Z = 1)$.
 4. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations.
 5. Compute $P(X + Z = 3)$.
 6. Suppose that Yolanda bets on black on all 5 spins of the wheel. Does $X + Y$ have a Binomial distribution? Explain.
 7. Suppose instead that Xavier bet on the number 7 in his two bets. Does $X + Z$ have a Binomial distribution in this case? Explain.
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- If X and Y are independent, X has a $\text{Binomial}(n_X, p)$ distribution, and Y has a $\text{Binomial}(n_Y, p)$ distribution then $X + Y$ has a $\text{Binomial}(n_X + n_Y, p)$
 - A $\text{Binomial}(1, p)$ distribution is also known as a **Bernoulli**(p) distribution, taking a value of 1 with probability p and 0 with probability $1 - p$. Any *indicator random variable* has a Bernoulli distribution.
 - If $X_1, X_2, \dots, X_n$ are independent each with a $\text{Bernoulli}(p)$ distribution, then $X_1 + \dots + X_n$ has a $\text{Binomial}(n, p)$ distribution.